Cofinality

Written by: xennonio

Summary

1 Motivation **3**

1.1 Classical Examples . . . 3

1.2 Definition of Cofinality . . . 4

1.3 Regular and Singular Cardinals . . . 4

2 Intuitions on Cofinality **6**

2.1 Distribution of Regular and Singular Cardinals . . . 6

2.2 Some Basic Properties . . . 7

2.3 Uncountable Limit Regular Cardinals . . . 8

2.4 Rambling on Cofinality . . . 9

2.5 Additional Properties of Cofinality . . . 9

3 Cardinals Exponentiation **11**

3.1 König's Theorem . . . 11

3.1.1 Cardinal Exponentiation for Regular Cardinals . . . 12

3.1.2 Digression on Exponentiation for Singular Cardinals . . . 13

3.2 Evaluating $\aleph_\alpha^{\aleph_\beta}$ . . . 14

3.2.1 Under GCH . . . 15

3.2.2 Hausdorff's Formula . . . 16

4 Silver's Theorem **18**

4.1 Clubs . . . 18

4.2 Stationary Sets and Fodor's Lemma . . . 19

4.2.1 Applications . . . 21

4.3 Silver's Theorem . . . 22

Preface

These notes aim to introduce the concept of cofinality, offering the motivations and intuitions behind it. The initial chapters consist primarily in results and examples that explore the idea of cofinality, showing how it naturally arises and examining its key properties.

From there on, almost everything presented can be found in Jech's book - Introduction to Set Theory, all I will do here is a comprehensive and filtered summary, along with some of my own intuitions that may be useful to someone, and some additional comments on things I've found elsewhere.

Notably, the concept of cofinality leads us directly to the Large Cardinal Axioms. We will explore how weakly inaccessible cardinals emerge by addressing a fundamental question in the study of regular and singular cardinals: "does there exist an uncountable regular limit cardinal?"

It is important to note that these notes assume the reader has some familiarity with the basics of cardinal and ordinal arithmetic to fully understand the proofs of the theorems. However, even without this background, readers can still grasp the underlying intuition by skipping the proofs and following the comments and ramblings throughout the text, which, hopefully, will at least guides you through a natural exploration of the defined concepts.

CHAPTER 1

Motivation

1.1 Classical Examples

Consider the following theorems:

- The finite union of finite sets is finite, i.e., if S is a system of sets such that $|S| < \omega_0$ and, for all $A \in S$, we have $|A| < \omega_0$, then

$$\left|\bigcup S\right| < \omega_0$$

- Similarly, the countable union of countable sets is countable, i.e., if S is a system of sets such that $|S| < \omega_1$ and, for all $A \in S$, we have $|A| < \omega_1$, then

$$\left|\bigcup S\right| < \omega_1$$

In both cases, the key takeaway is that it's difficult to "reach" $\aleph_0$ and $\aleph_1$, respectively, "from below". This phenomenon can be formally understood through the concept of *cofinality*.

To generalize, consider the following question: we know that any limit ordinal is the supremum of the set of all smaller ordinals. In particular, for any limit ordinal λ

$$\lambda = \bigcup \lambda$$

However, we don't necessarily need to take *all* smaller ordinals. It's possible to find a proper subset S of λ such that $\lambda = \sup(S)$. How small can S be? This is the concept we aim to explore.

1.2 Definition of Cofinality

Definition 1.1: Cofinality

We will define $\mathrm{cf}(\alpha)$, the cofinality of α, with α a limit ordinal, as the smallest ordinal ϑ such that α is the limit of an increasing transfinite sequence $\langle \alpha_\nu : \nu < \vartheta \rangle$ of length ϑ, i.e.

$$\alpha = \lim_{\nu \to \vartheta} \alpha_\nu$$

Equivalently, making easier to visualize the relation between this with the discussion above, $\mathrm{cf}(\alpha)$ is the smaller ordinal β such that exists $f : \beta \to \alpha$, satisfying

$$\alpha = \bigcup_{\nu < \beta} f(\nu) = \bigcup \mathrm{Im}(f)$$

i.e., $\mathrm{cf}(\alpha)$ is the smaller ordinal β such that there exists a union of β elements α_ν less than α.

1.3 Regular and Singular Cardinals

We can now define two types of cardinals, κ: those that can be written as a union of κ smaller elements, called singular, and those that cannot, called regular. More formally:

Definition 1.2: Regular and Singular Cardinal

An infinite cardinal κ is singular iff there exists an increasing transfinite sequence $\langle \alpha_\nu : \nu < \vartheta \rangle$ of ordinals $\alpha_\nu < \kappa$ where the length ϑ is a limit ordinal less than κ, such that

$$\kappa = \lim_{\nu \to \vartheta} \alpha_\nu$$

An infinite cardinal that is not singular is called regular.

Alternatively, we can define these cardinals more concretely using the following lemma:

Lemma 1.1

An infinite cardinal κ is singular if, and only if, it's the sum of fewer than κ smaller cardinals, i.e.

$$\kappa = \sum_{i \in I} \kappa_i$$

where $|I| < \kappa$ and $\kappa_i < \kappa$, for all $i \in I$.

Proof. ($\Rightarrow$) If κ is singular, then $\kappa = \lim_{\nu \to \vartheta} \alpha_\nu$ for some increasing transfinite sequence with $\alpha_\nu < \kappa$ review and $\vartheta < \kappa$. Since all ordinal is the set of all smaller ordinals, then

$$\kappa = \bigcup_{\nu < \vartheta} \alpha_\nu = \bigcup_{\nu < \vartheta} \left(\alpha_\nu \setminus \bigcup_{\xi < \nu} \alpha_\xi \right)$$

defining $A_\nu := \alpha_\nu \backslash \bigcup_{\xi<\nu} \alpha_\xi$, we have that $\langle A_\nu : \nu < \vartheta \rangle$ is a sequence of length less than κ with disjoint sets of cardinality $\kappa_\nu = |A_\nu| = \left|\alpha_\nu \backslash \bigcup_{\xi<\nu} \alpha_\xi\right| \leq |\alpha_\nu| < \kappa$, hence

$$\kappa = \sum_{\nu<\vartheta} \kappa_\nu$$

($\Leftarrow$) Let's assume that $\kappa = \sum_{\alpha<\lambda} \kappa_\alpha$, with $\lambda, \kappa_\alpha < \kappa$, for all $\alpha < \lambda$. Therefore

$$\kappa = \sum_{\alpha<\lambda} \kappa_\alpha = \lambda \cdot \sup_{\alpha<\lambda} \kappa_\alpha$$

and, since $\lambda < \kappa$, then necessarily $\kappa = \sup_{\alpha<\lambda} \kappa_\alpha$, because, otherwise, $\kappa = \lambda$, a contradiction. Then κ is the supremum of the image of the transfinite sequence $\langle \kappa_\alpha : \alpha < \lambda \rangle$, hence, it's easy to find an increasing subsequence with limit κ. ■

Note that such characterization captures well the idea discussed at motivation. We can relate it to the concept of cofinality as follows: it's expected and easy to verify that, if κ is a singular cardinal, then $\mathrm{cf}(\kappa) < \kappa$, i.e., there is an increasing sequence of length smaller than κ with elements smaller than κ, but which has κ as a limit and, if κ is regular, since $\mathrm{cf}(\kappa) \leq \kappa$, we have that $\mathrm{cf}(\kappa) = \kappa$.

CHAPTER 2

Intuitions on Cofinality

2.1 Distribution of Regular and Singular Cardinals

To facilitate notation, we will say that a cardinal $\aleph_\alpha$ is a successor cardinal if α is a successor ordinal, and we will say that it's a limit cardinal if α is a limit ordinal.

Let's explore the concept we create with some cardinals. For the cases used in the motivation, we have that $\mathrm{cf}(\aleph_0) = \aleph_0$ and $\mathrm{cf}(\aleph_1) = \aleph_1$, i.e., both are regular cardinals, since they can't be written as a union of $\aleph_0$ or $\aleph_1$ smaller elements, respectively. In particular, the following lemma states that every successor cardinal $\aleph_{\alpha+1}$ is a regular cardinal, that is, it's not easy to "reach them from below", only with smaller sets.

Lemma 2.1

Every successor cardinal $\aleph_{\alpha+1}$ is a regular cardinal.

Proof. Assume by contradiction that $\aleph_{\alpha+1}$ is singular, by lemma 1.1, it can be written as a sum of a smaller number of smaller cardinals:

$$\aleph_{\alpha+1} = \sum_{i \in I} \kappa_i$$

with $|I| < \aleph_{\alpha+1}$ and $\kappa_i < \aleph_{\alpha+1}$, $\forall i \in I$. Therefore $|I|, \kappa_i \leq \aleph_\alpha$, $\forall i \in I$, so, we have

$$\aleph_{\alpha+1} = \sum_{i \in I} \kappa_i \leq \sum_{i \in I} \aleph_\alpha = \aleph_\alpha \cdot |I| \leq \aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha$$

a contradiction, then $\aleph_{\alpha+1}$ is a regular cardinal. ∎

On the other hand, we have

$$\begin{aligned}
\aleph_\omega &= \lim_{n\to\omega} \aleph_n \\
\aleph_{\omega+\omega} &= \lim_{n\to\omega} \aleph_{\omega+n} \\
\aleph_{\omega\cdot\omega} &= \lim_{n\to\omega} \aleph_{\omega\cdot n} \\
\aleph_{\omega_1} &= \lim_{\alpha\to\omega_1} \aleph_\alpha
\end{aligned}$$

It's easy to see that the vast majority of limit cardinals we have contacts with, except for $\aleph_0$, are indeed singular; we will state this more precisely in the next theorem. Furthermore, each cardinal on the examples shown above are singular, the first three had cofinality ω and the last one cardinality ω_1. We will see how powerful this is when we present other intuitive ways of visualizing them, but for now let us answer some natural questions that arise, such as: are there arbitrarily large singular cardinals? Or rather, are all uncountable limit cardinals singular?

2.2 Some Basic Properties

Let's us first show that there do indeed exist arbitrarily large singular cardinals, and further, if $\aleph_\alpha$ is a singular limit cardinal, then the smallest limit cardinal larger than $\aleph_\alpha$, i.e., the next limit cardinal, is also singular.

Lemma 2.2

There are arbitrarily large singular cardinals.

Proof. Let $\aleph_\alpha$ be an arbitrary cardinal and consider the following sequence

$$\aleph_\alpha, \aleph_{\alpha+1}, \aleph_{\alpha+2}, \ldots, \aleph_{\alpha+n}, \ldots$$

For $n \in \omega$. Therefore

$$\aleph_{\alpha+\omega} = \lim_{n\to\omega} \aleph_{\alpha+n}$$

and, then, $\aleph_{\alpha+\omega}$ is a singular cardinal greater than $\aleph_\alpha$. ■

As a scholium of the previous lemma, we have

Corollary 2.1

If $\aleph_\alpha$ is a singular cardinal, then the next limit cardinal $\aleph_{\alpha+\omega}$ is also singular. In particular, if $\aleph_\alpha$ was regular, the next limit cardinal is also singular.

With this, we can have a visualization of which of the first cardinals are regular or singular. We have that, after $\aleph_0$, each successor cardinal below $\aleph_\omega$ is regular, and $\aleph_\omega$ is itself singular. We can repeat it from $\aleph_\omega$ to $\aleph_{\omega\cdot 2}$, $\aleph_{\omega^\omega}$, $\aleph_{\varepsilon_0}$, and so on. In fact, even reaching $\aleph_{\omega_1}$, which is singular, we will have the same between it and the next limit $\aleph_{\omega_1+\omega}$. So the "line" of cardinals has, apparently, almost always a singular cardinal in the limit points, and always countable many regular successor cardinals between them.

2.3 Uncountable Limit Regular Cardinals

Let's now investigate how would be an uncountable limit regular cardinal. Let $\aleph_\lambda$ be such cardinal, since λ is a limit ordinal, then

$$\aleph_\lambda = \lim_{\alpha \to \lambda} \aleph_\alpha$$

i.e., $\aleph_\lambda$ is the limit of an increasing sequence of length α, in particular, we could change such sequence to another one of length $\mathrm{cf}(\lambda)$. It's easy to see how this imply the next result

Lemma 2.3

If λ is a limit ordinal, then $\mathrm{cf}(\aleph_\lambda) = \mathrm{cf}(\lambda)$

Now, since in our case $\aleph_\lambda$ is also a regular cardinal, then $\aleph_\lambda = \mathrm{cf}(\aleph_\lambda) = \mathrm{cf}(\lambda) \leq \lambda$, which imply

$$\boxed{\aleph_\lambda = \lambda}$$

Such cardinals are called aleph fixed points, since, well, they're a fixed point of the aleph operation.

Now, we already know $\aleph_\omega \neq \omega$, and $\aleph_{\omega_1} \neq \omega_1$, $\aleph_{\omega_2} \neq \omega_2$, etc. and the gap between the indices ω_1 and 1, ω_2 and 2, and so on keep increasing! Similarly, we have $\aleph_{\aleph_\omega} \neq \aleph_\omega$. Repeating this arguments it's easy to see that such finite towers of $\aleph$'s cannot reach an aleph fixed point. In fact, it shows that the first aleph fixed point is of the form

$$\aleph_{\aleph_{\aleph_{\ddots}}} = \bigcup_{n \in \omega} \aleph_{\aleph_{\aleph_{\ddots_{\aleph_0}}}}$$

where we have n $\aleph$'s on the RHS.

Such property and example suggest these cardinals should be really big. However, the next Lemma reveals us that, in fact, this property isn't too strong as it seems, since we can actually prove that there are arbitrarily large singular aleph fixed points, as the next lemma[1] states.

add footnote hyperlink

Lemma 2.4

There are arbitrarily large singular aleph fixed points.

Proof. Let κ be an arbitrary cardinal and consider the following sequence α_n defined recursively as $\alpha_0 = \aleph_\gamma$, $\alpha_{n+1} = \aleph_{\alpha_n}$, for each $n \in \omega$. Now, assume by contradiction that there is not singular $\alpha \geq \kappa$ which is a aleph fixed point, then α_n is an increasing sequence; let $\alpha := \lim_{n \to \omega} \alpha_n$. Obviously the sequence $\aleph_{\alpha_n} = \alpha_{n+1}$ also has limit $\aleph_\alpha$, therefore, we have that

$$\aleph_\alpha = \lim_{n \to \omega} \aleph_{\alpha_n} = \lim_{n \to \omega} \alpha_{n+1} = \alpha$$

Then $\alpha = \aleph_\alpha$ is a fixed aleph point which is also a limit of an increasing sequence of length ω, i.e., $\mathrm{cf}(\alpha) = \omega < \alpha$, then it's singular. ■

[1] In fact, this lemma is the toy version of a more general fact known as The Fixed-Point Lemma for normal functions, which guarantees us that every normal function has arbitrarily large fixed points. You can see this in its full generality on my Large Cardinals notes, the proof follows the same idea as the one used to prove the lemma.

The next natural step is then ask what would be an example of an uncountable regular limit cardinal. Well, surprisingly, such cardinals, named weakly inaccessible cardinals, are so big that ZFC can't prove they exist! Inaccessible cardinals are some of the first cardinals in a larger class of properties named Large Cardinals Properties.

Unfortunatelly, the proof is not that easy, since we need to use a consistent result, namely, that GCH is consistent with ZFC[2]. However, we can prove it for a somewhat similar cardinal, named strongly limit cardinals, you can see how to prove it in the Large Cardinals notes, and then how to use it to show ZFC also can't prove weakly inaccessible exist.

add hyperlink

2.4 Rambling on Cofinality

Now we already motivated Large Cardinal, let's ramble a little about cofinality. Some books, before defining cofinality, use the following definitions: a set $C \subseteq \lambda$ is cofinal or unbounded in λ if for all $\alpha < \lambda$, there exists a $\gamma \in C$ such that $\alpha < \gamma$. It's not hard to see that this is equivalent to $\bigcup C = \lambda$.

We can then define $\mathrm{cf}(\lambda)$ as the least cardinality of an unbounded subset of λ. In other words, what we are saying is that subsets of regular cardinal need almost all elements to become unbounded in it, whereas singular cardinals don't.

Now, imagine we have some property about ordinals smaller than $\aleph_\omega$, and such property is somewhat inductive. Then the process of extending this property to $\aleph_\omega$ itself would require only ω many steps, since $\mathrm{cf}(\aleph_\omega) = \omega$, whereas, for example, doing the same for $\aleph_1$ would require uncountable many steps.

Of course none of this is really formal, but you will get your own intuition about this while getting deeper and deeper in set theory. You will see that a lot of theorems states something only for regular or only for singular cardinals, and that they influence a lot, for example, in cardinal arithmetic. In fact, we will see in the next chapters that cardinal arithmetic depends a lot of when the cardinals we are using are either singular or regular.

2.5 Additional Properties of Cofinality

Let's finish this section with some extra results about cofinality.

We already proved that if $\aleph_\alpha$ is singular, then $\mathrm{cf}(\omega_\alpha) < \omega_\alpha$ and, if it's regular, then $\mathrm{cf}(\omega_\alpha) = \omega_\alpha$. We will now show that

Lemma 2.5

If a limit ordinal α isn't a cardinal, then $\mathrm{cf}(\alpha) < \alpha$. In particular, for all limit ordinal α, we have $\mathrm{cf}(\alpha) = \alpha$ if, and only if, α is a regular cardinal.

Proof. Let α be an ordinal which is not a cardinal, then $\kappa = |\alpha| < \alpha$ and there is a bijection $f : \kappa \to \alpha$, i.e., a sequence $\langle \alpha_\nu = f(\nu) : \nu < \kappa \rangle$ of length κ such that $\{\alpha_\nu : \nu < \kappa\} = \alpha$.

Now, even if such sequence was not increasing, we could find, using transfinite induction, an increasing subsequence with limit α. And, since the length of the subsequence is at most $\kappa < \alpha$, then $\mathrm{cf}(\alpha) < \alpha$. ■

[2] You can find a proof of it in my Consistency Results notes, in Section 3. The Constructible Universe L.

Lemma 2.6

For all limit ordinal α, we have

$$\mathrm{cf}(\mathrm{cf}(\alpha)) = \mathrm{cf}(\alpha)$$

Proof. Let $\vartheta = \mathrm{cf}(\alpha)$. Clearly ϑ is a limit ordinal and we already know that $\mathrm{cf}(\vartheta) \leq \vartheta$. Now, assume by a contradiction that $\gamma = \mathrm{cf}(\vartheta) < \vartheta$, then there is an increasing sequence $\langle \nu_\xi : \xi < \gamma \rangle$ with $\lim_{\xi \to \gamma} \nu_\xi = \vartheta$. But since $\vartheta = \mathrm{cf}(\alpha)$, there is also a $\langle \alpha_{\nu_\xi} : \xi < \gamma \rangle$ of length $\gamma < \vartheta$ with limit α, contradicting the minimality of ϑ. ■

Corollary 2.2

For all limit ordinal α, $\mathrm{cf}(\alpha)$ is a regular cardinal.

CHAPTER 3

Cardinals Exponentiation

While addition and multiplication of cardinals is very simple, since

$$\aleph_\alpha + \aleph_\beta = \aleph_\alpha \cdot \aleph_\beta = \max\{\aleph_\alpha, \aleph_\beta\}$$

exponentiation is way more complex.

In particular, we will see that The Generalized Continuum Hypothesis (GCH):

$$2^{\aleph_\alpha} = \aleph_{\alpha_+1}, \quad \text{for all } \alpha$$

simplify cardinal exponentiation enormously and, without it, for regular cardinals there isn't much to prove[1] besides

$$2^{\aleph_\alpha} \geq \aleph_{\alpha+1}, \qquad \alpha \leq \beta \Rightarrow 2^{\aleph_\alpha} \leq 2^{\aleph_\beta} \tag{1}$$

and the next result about cofinality.

3.1 König's Theorem

Before proving some theorems in cardinal exponentiation, let's prove one of the most famous results in cardinal arithmetic, known as König's Theorem, which is a strict inequality! Notice how rare this is, the only other strict inequality we saw in cardinal arithmetic was probably Cantor's Theorem. In fact, we will see Cantor is a direct consequence of König's.

Theorem 3.1: König's Theorem

If $\kappa_i < \lambda_i$, where κ_i and λ_i are cardinal, $\forall i \in I$, then

$$\sum_{i \in I} \kappa_i < \prod_{i \in I} \lambda_i$$

[1] in fact, this is a statement in the formal sense. We have a lot of results of the form: giving some relations between cardinals, any possible assignment to those cardinals that doesn't contradicts such relations is realized in some model.

The idea behind the proof is: If $\kappa_i < \lambda_i$, then every function $f_i : \kappa_i \to \lambda_i$ isn't surjective, therefore, we can find an element $b_i \in \lambda_i \backslash \mathrm{Im}(f_i)$, and then we obtain, for each function $f : \bigcup_{i\in I} A_i \to \prod_{i\in I} B_i$, the element $b = \langle b_i : i \in I\rangle$ which is not in the image of f.

Proof. Let $\langle A_i : i \in I\rangle$ and $\langle B_i : i \in I\rangle$ be such that $|A_i| = \kappa_i$ and $|B_i| = \lambda_i$, with A_i mutually disjoint, $\forall i \in I$. Since $|A_i| < |B_i|$, then, given $f_i : A_i \to B_i$ injective, we know they can't be injective. So, define

should I use daggers?

$$B_i^\dagger = \{x \in B_i : x \notin f_i[A_i]\}$$

Since f_i isn't surjective, $B_i^\dagger \neq \emptyset$, $\forall i \in I$, then, by AC, we can choose $b_i^\dagger \in B_i^\dagger$, $\forall i \in I$. Now, let

$$F : \bigcup_{i\in I} A_i \to \prod_{i\in I} B_i$$

be the unique function satisfying

$$\pi_j(F(a)) = \begin{cases} f_i(a), & i = j \\ b_j^\dagger, & i \neq j \end{cases}$$

Now, if $F(x) = F(y)$, with $x \in A_i$ and $y \in A_j$, then

$$\pi_i(F(x)) = \pi_i(F(y))$$

if $i \neq j$, then

$$\pi_i(F(x)) = f_i(x) = b_i^\dagger = \pi_i(F(y))$$

a contradiction, since $b_i^\dagger \notin f_i[A_i]$. If $i = j$, then $f_i(x) = f_i(y)$, but, since f_i is injective, $x = y$. This proves F is an injection. Let's now prove there is no surjection.

Assume by contradiction that there is some $G : \bigcup_{i\in I} A_i \to \prod_{i\in I} B_i$ surjective, and define

$$B_i^\ddagger = \{x \in B_i : x \notin \pi_i(G[A_i])\}$$

Since $(\pi_i \circ G) : A_i \to B_i$, then it's not surjective and, therefore, $B_i^\ddagger \neq \emptyset$, $\forall i \in I$. By AC there is a $b = \left\langle b_i^\ddagger : i \in I\right\rangle$ such that $b_i^\ddagger \in B_i^\ddagger$. Since G is surjective, there is a $x \in A_i$, for some $i \in I$, such that $G(x) = b^\ddagger$, so $\pi_i(G(x)) = \pi_i\left(b^\ddagger\right) \in B_i^\ddagger$, a contradiction, then G is not surjective. ■

Notice that the choices of $B_i^\dagger$ and $B_i^\ddagger$ in the proof looks like the choice of $X = \{x \in A : x \notin f(x)\}$ in Cantor's Theorem proof and, in fact, Cantor's Theorem is a corollary of König's Theorem for $\kappa_i = 1$ and $\lambda_i = 2$, $\forall i \in I$.

3.1.1 Cardinal Exponentiation for Regular Cardinals

With such big results in hands, we can prove the following lemma

Lemma 3.1

For any ordinal α

$$\mathrm{cf}\left(2^{\aleph_\alpha}\right) > \aleph_\alpha$$

Note that this lemma restricts the possible values $2^{\aleph_\alpha}$ can assume, depending on its cofinality. As an example, which also reinforce the comments in section 2.4 and in the last footnote, notice that we can't have $2^{\aleph_0} = \aleph_\omega$, since then we would have $\aleph_0 = \mathrm{cf}\,(\aleph_\omega) = \mathrm{cf}\left(2^{\aleph_0}\right) > \aleph_0$, a contradiction.

In fact, Robert Solovay published a result in a really famous one page paper named "$2^{\aleph_0}$ can be anything it ought to be" which basically says that the only two things we can prove about the continuum is: that it's uncountable, and that it has uncountable cofinality. Formally, the result says that for any uncountable cardinal κ with uncountable cofinality, there is a model of ZFC where this is realized, i.e., where $2^{\aleph_0} = \kappa$.

Nowadays, such result is a direct consequence of a more general one known as Easton's Theorem, that says basically the same thing, but for any regular cardinal, i.e., the only constraints on permissible values for 2^κ when κ is a regular cardinal are eq. (1) and the previous lemma.

Proof. Let $\vartheta = \mathrm{cf}\left(2^{\aleph_\alpha}\right)$, then, there are $\kappa_\nu < 2^{\aleph_\alpha}$, $\nu < \vartheta$ such that

$$2^{\aleph_\alpha} = \sum_{\nu<\vartheta} \kappa_\nu$$

Using König's Theorem for $\lambda_\nu = 2^{\aleph_\alpha}$, we have

$$2^{\aleph_\alpha} = \sum_{\nu<\vartheta} \kappa_\nu < \prod_{\nu<\vartheta} 2^{\aleph_\alpha} = \left(2^{\aleph_\alpha}\right)^\vartheta$$

Now, assume by contradiction that $\vartheta \leq \aleph_\alpha$, then

$$2^{\aleph_\alpha} > \left(2^{\aleph_\alpha}\right)^\vartheta \leq \left(2^{\aleph_\alpha}\right)^{\aleph_\alpha} = 2^{\aleph_\alpha \cdot \aleph_\alpha} = 2^{\aleph_\alpha}$$

a contradiction, so $\vartheta = \mathrm{cf}\left(2^{\aleph_\alpha}\right) > \aleph_\alpha$. ■

Notice how we didn't use anything special about 2, and then we can repeat the exact same argument for any cardinal $\kappa > 1$, giving us:

Corollary 3.1

For any cardinals $\kappa, \lambda > 1$, we have

$$\mathrm{cf}\left(\kappa^\lambda\right) > \lambda$$

3.1.2 Digression on Exponentiation for Singular Cardinals

As we saw, the inequalities above are the only ones we can prove about regular cardinals, the next natural step is to ask if there is a generalization, or some similar result about singular cardinals. In fact, for singular cardinals κ, various additional constraints in the behavior of 2^κ are known.

By know, the most interesting one is Silver's Theorem, since it implies that Easton's Theorem can't be extended to include singular cardinals. But, nowadays, Shelah's work on PCF Theory gives a lot of results on the possible values of 2^κ, for κ singular. It shows that such values are strongly influenced by the values on smaller cardinals, as we will see next and when proving Silver's Theorem, whereas Easton shows this is not the case for regular cardinals.

Lemma 3.2

Let $\aleph_\alpha$ be a singular cardinal, if $2^{\aleph_\xi} = \aleph_\beta$ for all $\xi < \alpha$, then $2^{\aleph_\alpha} = \aleph_\beta$.

Proof. Since $\aleph_\alpha$ is singular, there are $\kappa_i < \aleph_\alpha$, $i \in I$, with $|I| = \aleph_\lambda < \aleph_\alpha$ such that

$$2^{\aleph_\alpha} = \sum_{i \in I} \kappa_i$$

By hypothesis $2^{\kappa_i} = 2^{\aleph_\lambda} = \aleph_\beta$, then

$$2^{\aleph_\alpha} = 2^{\sum_{i \in I} \kappa_i} = \prod_{i \in I} 2^{\kappa_i} = \prod_{i \in I} \aleph_\beta = (\aleph_\beta)^{\aleph_\lambda} = \left(2^{\aleph_\lambda}\right)^{\aleph_\lambda} = 2^{\aleph_\lambda} = \aleph_\beta$$

■

3.2 Evaluating $\aleph_\alpha^{\aleph_\beta}$

Let's first state some important lemmas. The first simplify $\aleph_\alpha^{\aleph_\beta}$ for when $\alpha \leq \beta$.

Lemma 3.3

If $\alpha \leq \beta$, then

$$\aleph_\alpha^{\aleph_\beta} = 2^{\aleph_\beta}$$

Proof. Since $2 < \aleph_\alpha$, we have $2^{\aleph_\beta} \leq \aleph_\alpha^{\aleph_\beta}$. By Cantor's Theorem $\aleph_\alpha < 2^{\aleph_\alpha}$, then

$$\aleph_\alpha{}^{\aleph_\beta} \leq \left(2^{\aleph_\alpha}\right)^{\aleph_\beta} = 2^{\aleph_\alpha \cdot \aleph_\beta} = 2^{\aleph_\beta}$$

■

The next natural step is trying to calculate $\aleph_\alpha^{\aleph_\beta}$ for when $\alpha > \beta$. The next lemma will show itself useful when trying to do this.

Lemma 3.4

If $\alpha \geq \beta$, then

$$\left|[\omega_\alpha]^{\aleph_\beta}\right| = \aleph_\alpha^{\aleph_\beta}$$

Proof. Since $|\omega_\alpha \times \omega_\beta| = \aleph_\alpha \cdot \aleph_\beta = \aleph_\alpha = |\omega_\alpha|$, then $\left|[\omega_\alpha \times \omega_\beta]^{\aleph_\beta}\right| = \left|[\omega_\alpha]^{\aleph_\beta}\right|$. Now, every function review $f : \omega_\beta \to \omega_\alpha$ is in $[\omega_\alpha \times \omega_\beta]^{\aleph_\beta}$, therefore $\omega_\alpha^{\omega_\beta} \subseteq [\omega_\alpha \times \omega_\beta]^{\aleph_\beta}$ and so

$$\left|\omega_\alpha^{\omega_\beta}\right| = \aleph_\alpha^{\aleph_\beta} \leq \left|[\omega_\alpha \times \omega_\beta]^{\aleph_\beta}\right| = \left|[\omega_\alpha]^{\aleph_\beta}\right|$$

For the reverse inequality, if $X \in [\omega_\alpha]^{\aleph_\beta}$, then there is some f in ω_b such that $\mathrm{Im}(f) = X$. With this, given a function f_X for each $X \in [\omega_\alpha]^{\aleph_\beta}$, let $F(X) = f_X$. If $X \neq Y$, then $\mathrm{Im}(f_X) = X$ and $\mathrm{Im}(f_Y) = Y$, so $F(X) = f_X \neq f_Y = F(y)$, i.e., F is injective, then

$$\left|[\omega_\alpha]^{\aleph_\beta}\right| \leq \left|\omega_\alpha^{\omega_\beta}\right| = \aleph_\alpha^{\aleph_\beta}$$

■

3.2.1 Under GCH

Now, as a warmup, let's see what we got under the Generalized Continuum Hypothesis (GCH).

Lemma 3.5: (Under GCH)

If $\aleph_\alpha$ is a regular cardinal, then

$$\aleph_\alpha^{\aleph_\beta} = \begin{cases} \aleph_\alpha, & \text{if } \alpha > \beta \\ \aleph_{\beta+1}, & \text{if } \alpha \leq \beta \end{cases}$$

Proof. If $\alpha \leq \beta$, by lemma 3.3 $\aleph_\alpha^{\aleph_\beta} = 2^{\aleph_\beta} = \aleph_{\beta+1}$. Let then $\alpha > \beta$. If we let $S = [\omega_\alpha]^{\aleph_\beta}$, by lemma 3.3 we have $|S| = \aleph_\alpha^{\aleph_\beta}$. In section 2.3, we saw $\mathrm{cf}(\kappa)$ can be interpreted as the least size of an unbounded set in κ. In particular, since $\mathrm{cf}(\aleph_\alpha) = \aleph_\alpha$ for regular cardinals, we know every $X \subseteq \omega_\alpha$, with $|X| < \aleph_\alpha$, is limited. Let then review

$$B = \bigcup_{\delta < \omega_\alpha} \mathcal{P}(\delta)$$

be the collection of all limited subsets of ω_α. We claim $|B| \leq \aleph_\alpha$; then, since $S \subset B$, we have $|S| = \aleph_\alpha^{\aleph_\beta} = \aleph_\alpha$, since clearly $|S| = \aleph_\alpha^{\aleph_\beta} \geq \aleph_\alpha$. From the definition of B we have

$$|B| \leq \sum_{\delta < \omega_\alpha} 2^{|\delta|}$$

However, for any cardinal $\aleph_\gamma < \aleph_\alpha$, we also have $2^{\aleph_\gamma} = \aleph_{\gamma+1} \leq \aleph_\alpha$, therefore $2^{|\delta|} \leq \aleph_\alpha$, for any $\delta < \omega_\alpha$, then

$$|B| \leq \sum_{\delta < \omega_\alpha} 2^{|\delta|} \leq \sum_{\delta < \omega\alpha} \aleph_\alpha = \aleph_\alpha \cdot \aleph_\alpha = \aleph_\alpha$$

■

Let's now prove a similar, but more complicated, formula for singular cardinals.

Lemma 3.6: (Under GCH)

If $\aleph_\alpha$ is a singular cardinal, then

$$\aleph_\alpha^{\aleph_\beta} = \begin{cases} \aleph_\alpha, & \text{if } \aleph_\beta < \mathrm{cf}\,(\aleph_\alpha), \\ \aleph_{\alpha+1}, & \text{if } \mathrm{cf}\,(\aleph_\alpha) \leq \aleph_\beta \leq \aleph_\alpha, \\ \aleph_{\beta+1}, & \text{if } \aleph_\beta \geq \aleph_\alpha \end{cases}$$

Schematically, we obtain fig. 3.1.

Proof. If $\beta \geq \alpha$, then $\aleph_\beta \geq \aleph_\alpha$ and $\aleph_\alpha^{\aleph_\beta} = 2^{\aleph_\beta} = \aleph_{\beta+1}$. If $\aleph_\beta < \mathrm{cf}\,(\aleph_\alpha)$, then any $X \subseteq \omega_\alpha$ with $|X| = \aleph_\beta < \mathrm{cf}\,(\aleph_\alpha) < \aleph_\alpha$ is bounded in ω_α and, therefore, by the same argument as in the previous theorem, we have $\aleph_\alpha^{\aleph_\beta} = \aleph_\alpha$. Let then $\mathrm{cf}\,(\aleph_\alpha) \leq \aleph_\beta \leq \aleph_\alpha$. On the one hand review

$$\aleph_\alpha \leq \aleph_\alpha^{\aleph_\beta} \leq \aleph_\alpha^{\aleph_\alpha} = 2^{\aleph_\alpha} = \aleph_{\alpha+1} \tag{2}$$

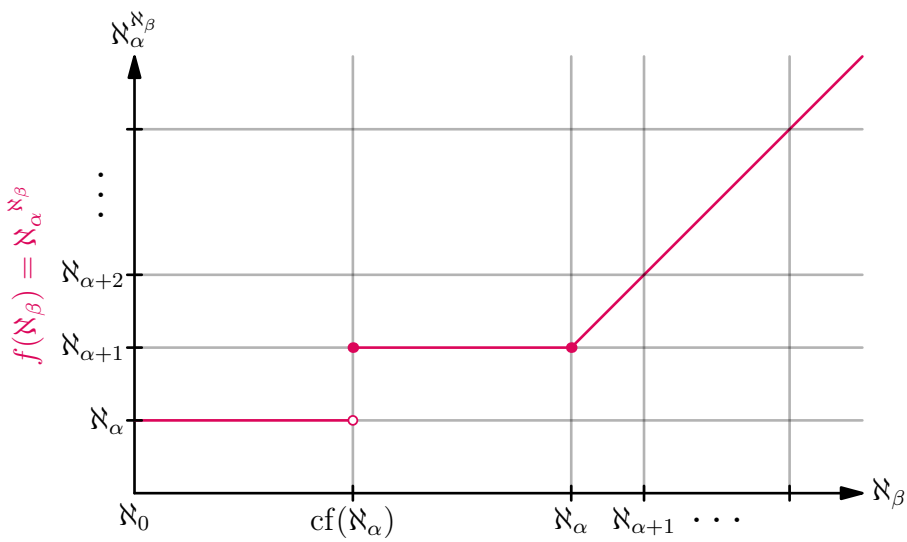

Fig. 3.1: graph of $f(\aleph_\beta) = \aleph_\alpha^{\aleph_\beta}$ under GCH, for $\aleph_\alpha$ singular.

But on the other, by corollary 3.1, $\mathrm{cf}\left(\aleph_\alpha^{\aleph_\beta}\right) > \aleph_\beta$ and, since $\aleph_\beta \geq \mathrm{cf}\,(\aleph_\alpha)$, then $\mathrm{cf}\left(\aleph_\alpha^{\aleph_\beta}\right) \neq \mathrm{cf}\,(\aleph_\alpha)$, so $\aleph_\alpha^{\aleph_\beta} \neq \aleph_\alpha$. With this and eq. (2), the only possible value for $\aleph_\alpha^{\aleph_\beta}$ is $\aleph_{\alpha+1}$. ■

3.2.2 Hausdorff's Formula

Lastly, let's see what we can do without GCH.

Theorem 3.2: Hausdorff's Formula

For any ordinals α and β, we have

$$\aleph_{\alpha+1}^{\aleph_\beta} = \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$$

Proof. If $\beta \geq \alpha + 1$, then, since $\aleph_{\alpha+1} \leq \aleph_\beta \leq 2^{\aleph_\beta}$, we have review

$$\aleph_{\alpha+1}^{\aleph_\beta} = 2^{\aleph_\beta} = 2^{\aleph_\beta} \cdot \aleph_{\alpha+1} = \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$$

Let then $\beta \leq \alpha$. Since $\aleph_\alpha^{\aleph_\beta} \leq \aleph_{\alpha+1}^{\aleph_\beta}$ and $\aleph_{\alpha+1} \leq \aleph_{\alpha+1}^{\aleph_\beta}$, then

$$\aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1} \leq \aleph_{\alpha+1}^{\aleph_\beta} \cdot \aleph_{\alpha+1}^{\aleph_\beta} = \aleph_{\alpha+1}^{\aleph_\beta}$$

Therefore, it's sufficient to show $\aleph_{\alpha+1}^{\aleph_\beta} \leq \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$. Since $\omega_\beta < \omega_{\alpha+1}$ and $\omega_{\alpha+1}$ is regular, then for any $f : \omega_\beta \to \omega_{\alpha+1}$, f is bounded, i.e., there is some $\gamma < \omega_{\alpha+1}$ such that $f(\xi) < \gamma$, for every $\xi < \omega_\beta$. With this

$$\omega_{\alpha+1}^{\omega_\beta} = \bigcup_{\gamma < \omega_{\alpha+1}} \gamma^{\omega_\beta}$$

Now, any $\gamma < \omega_{\alpha+1}$ is such that $|\gamma| \leq \aleph_\alpha$ and, since

$$\left| \bigcup_{\gamma < \omega_{\alpha+1}} \gamma^{\omega_\beta} \right| \leq \sum_{\gamma < \omega_{\alpha+1}} |\gamma|^{\aleph_\beta}$$

then

$$\aleph_{\alpha+1}^{\aleph_\beta} \leq \sum_{\gamma<\omega_{\alpha+1}} |\gamma|^{\aleph_\beta} \leq \sum_{\gamma<\omega_{\alpha+1}} \aleph_\alpha^{\aleph_\beta} = \aleph_\alpha^{\aleph_\beta} \cdot \aleph_{\alpha+1}$$

■

CHAPTER 4

Silver's Theorem

In order to prove Silver's Theorem, we will need a really powerful concept called stationary sets. Besides Silver's Theorem, we will also see stationary sets proves a lot of other important results.

4.1 Clubs

Before defining stationary sets, let's first analyze a similar concept.

Throughout this section, κ will denote a regular cardinal.

Definition 4.1: Club

A set $C \subseteq \kappa$ is called a closed unbounded set in κ (or, more shortly, a club), if

- C is unbounded in κ;
- C is closed, i.e., $\sup(C \cap \alpha) \in C$ for all $\alpha < \kappa$.

Note that C being closed in κ is equivalent to say that every sequence in C of length $\alpha < \kappa$ has its limit in C. With this in mind, let's now see that clubs has the finite intersection property:

Lemma 4.1

If C_1 and C_2 are clubs, then $C_1 \cap C_2$ is also a club.

Proof. To see it's closed, let $\alpha < \kappa$. Since $\sup(C_i \cap \alpha) \in C_i$, for $i = 1, 2$, then

$$\sup((C_1 \cap C_2) \cap \alpha) = \bigcup((C_1 \cap \alpha) \cap (C_2 \cap \alpha)) = \left(\bigcup(C_1 \cap \alpha)\right) \cap \left(\bigcup(C_2 \cap \alpha)\right) \in C_1 \cap C_2$$

For unboundedness, let $\gamma < \kappa$, and consider the increasing sequence

$$\gamma < \alpha_0 < \beta_0 < \alpha_1 < \beta_1 < \cdots < \kappa$$

where $\alpha_n \in C_1$ and $\beta_n \in C_2$, for $n \in \omega$. In fact, they exists, since C_1 and C_2 are unbounded, and their supremum δ is the same. Now, since C_1 and C_2 are also closed, we know $\delta \in C_1 \cap C_2$. ■

The previous lemma guarantees us the set of all clubs generates a filter, called the club filter on κ:

$$F = \{X \subseteq \kappa : X \supseteq C \text{ for some club } C\}$$

In fact, such filter is κ-complete, which follows from the next lemma

Lemma 4.2

If $\{C_\alpha\}_{\alpha<\lambda}$ is a family of clubs, with $\lambda < \kappa$, then $\bigcap_{\alpha<\lambda} C_\alpha$ is also a club.

Proof. The idea of the proof is similar to the one for two clubs.

That it's closed is as easy as in the previous lemma. Given $\gamma < \kappa$, we have

$$\bigcup\left(\bigcap_{\alpha<\lambda}(C_\alpha \cap \gamma)\right) = \bigcap_{\alpha<\lambda}\left(\bigcup(C_\alpha \cap \gamma)\right) \in \bigcap_{\alpha<\lambda} C_\alpha$$

For unboundedness we can also try to follow a similar approach. In fact, as a warmup, consider $\kappa = \omega_1$ and $\lambda = \omega$. Then, given $\gamma < \omega_1$ we could consider the sequence

$$\gamma < \alpha_0^0 < \alpha_1^0 < \alpha_1^1 < \alpha_2^0 < \alpha_2^1 < \alpha_2^2 < \alpha_3^0 < \cdots < \kappa$$

where, for each $m \in \omega$, we have $\{\alpha_n^m\}_{n\in\omega} \subseteq C_m$.

Now, in order to generalize this process, given $\gamma < \kappa$, let $\beta_0 = \gamma$. Recursively, if β_n is already defined, since each C_α is unbounded, we can find some $\delta_{\alpha,n} > \beta_n$ in C_α. Let then $\beta_{n+1} = \sup_{\alpha<\lambda} \delta_{\alpha,n}$, which is a sequence of length $<\kappa$ with cardinals $<\kappa$. Since κ is regular, $\beta_{n+1} < \kappa$, and then we can repeat the process, obtaining $(\beta_n)_{n\in\omega}$. Finally, let $\beta = \sup_{n\in\omega} \beta_n$, again, by regularity of κ we have $\gamma < \beta < \kappa$ and since between each β_n, we have at least one element of each C_α, it's clear that β is also the supremum of each sequence $\delta_{\alpha,n}$ in C_α, which is closed, so $\beta \in C_\alpha$ for each $\alpha < \lambda$, then

$$\beta \in \bigcap_{\alpha<\lambda} C_\alpha$$

as desired. ■

4.2 Stationary Sets and Fodor's Lemma

Definition 4.2: Stationary Set

A set $S \subseteq \kappa$ is stationary if for every club C, we have $S \cap C \neq \emptyset$.

By the previous lemmas, every club is stationary; and if S is stationary, then any T such that $S \subseteq T \subseteq \kappa$ is also stationary. Interestingly, nonstationary sets are exactly the ones belonging to the ideal dual to the club filter.

Definition 4.3: Regressive Function

A function f with domain $S \subseteq \kappa$ is regressive if $f(\alpha) < \alpha$ for all $\alpha \neq 0$.

The next theorem[1] shows how uncountable regular cardinals differs substantially from $\aleph_0$. For example, it shows that there is no analogue for ω_1 of the function $f : \omega \to \omega$ defined by $f(n) = n-1$ for $n > 0$, and $f(0) = 0$, which is regressive, and assume each value in ω only finitely many times.

You can to try to define it using Cantor Normal Form. For successor ordinals $\alpha + 1$ it's easy, you can just let $f(\alpha+1) = \alpha$, but for limit ordinals, if you try, you will see you can't easily choose some smaller ordinal to be the image of f without making it also the image of a lot of other ordinals.

You can't, for example, choose $f(\alpha)$, for α limit, to be "the previous limit ordinal", since, for example, $\alpha = \omega^2$ doesn't have one. You also can't just crop it by removing the last term in Cantor Normal Form, since then all ordinals of the form ω^δ will be sent to 0.

Anyway, it's a nice exercise to try.

Theorem 4.1: Fodor's Lemma

A set $S \subseteq \kappa$ is stationary iff every regressive function $f : S \to \kappa$ is constant on an unbounded set. In fact, it will be constant in a stationary set.

Before proving it, we need an auxiliary lemma

Lemma 4.3

If $\{C_\xi\}_{\xi<\kappa}$ is a family of clubs, then the set $C \subseteq \kappa$ defined by

$$\alpha \in C \iff \alpha \in C_\xi,\ \forall \xi < \alpha$$

called the Diagonal Intersection of the C_ξ, is also a club.

Proof. To see it's closed, let $\lambda < \kappa$ and $(\alpha_\beta)_{\beta<\lambda}$ be an increasing sequence in C. Now, if $\xi < \alpha = \sup\{\alpha_\beta\}_{\beta<\lambda}$, by definition $\xi < \alpha_\delta$ for some $\delta < \lambda$. But then $\xi < \alpha_\beta$, for all $\beta \geq \delta$. Since each $\alpha_\beta \in C$, then $\alpha_\beta \in C_\xi$, for all $\beta \geq \delta$, but since C_ξ is closed, $\alpha = \sup\{\alpha_\beta : \beta \geq \delta\} \in C_\xi$ too. So we proved that $\alpha \in C_\xi$ for each $\xi < \alpha$, i.e., $\alpha \in C$.

For unboundedness, fix $\gamma < \kappa$, let's construct an increasing sequence $(\alpha_n)_{n\in\omega}$ as recursively as follows: let $\alpha_0 = \gamma$ and, if $\alpha_n < \kappa$ is already defined, we know by lemma 4.2 that $\bigcap_{\xi<\alpha_0} C_\xi$ is a club, then we can let α_{n+1} be the least element in it greater than α_n.

Now, let $\alpha = \sup_{n\in\omega} \alpha_n$, we claim $\alpha \in C$. In fact, if $\xi < \alpha$, by definition $\xi < \alpha_k$ for some $k \in \omega$, but then, by construction, $\alpha_{n+1} \in \bigcap_{\beta<\alpha_n} C_\beta$, in particular, $\alpha_{n+1} \in C_\xi$, for all $n \geq k$. ■

We can now prove Fodor's Lemma.

Proof. ($\Leftarrow$) Let $S \subseteq \kappa$ be nonstationary, it's sufficient to exhibit an example of a regressive $f : S \to \kappa$ that takes each value $<\kappa$-many times. Since S is nonstationary, there is some club C disjoint from S. Let then, for each $\alpha \in \kappa$, f be define as follows:

$$f(\alpha) = \sup(C \cap \alpha)$$

Since $f(\alpha) \in C$, $\alpha \in S$, and $S \cap C = \emptyset$, then $f(\alpha) < \alpha$. However, given any $\gamma < \kappa$, if δ is the least element of C greater than γ, then for any $\alpha > \delta$ we have

$$f(\alpha) = \sup(C \cap \alpha) \geq \delta > \gamma$$

[1] Which is usually called lemma

and then it can't be constant in γ on an unbounded set.

($\Rightarrow$) For each $\xi < \kappa$, let $S_\xi = f^{-1}(\xi)$, we will show S_ξ is stationary for some ξ.

Otherwise, for each $\xi < \kappa$ there would be a club C_ξ disjoint from S_ξ. Let C be the diagonal intersection of the C_ξ. Then $\alpha \in C$ iff $\alpha \in C_\xi$ for all $\xi < \alpha$, i.e., $f(\alpha) \neq \xi$ for all $\xi < \alpha$, this basically says that $f(\alpha) \geq \alpha$. But, by the previous Lemma, C is a club, hence it intersects S; however, f is regressive in S, a contradiction. ■

4.2.1 Applications

Let's now see how powerful Fodor's Lemma is.

First note that, until now, we saw every club is stationary, and then that every superset of a club is also stationary. However, we didn't see any example of a stationary set that doesn't contain a club. The next corollary will show us we can in fact find one[2]

Corollary 4.1

There are arbitrarily large stationary sets whose complement is also stationary.

Proof. Let C be the set of all limit ordinals below κ. For each $\alpha \in C$, there exists an increasing sequence $(x_{\alpha,n}) \to \alpha$. For each $n \in \omega$, let $f_n : C \to \kappa$ be given by $f_n(\alpha) = x_{\alpha,n}$.

Since $f_n(\alpha) < \alpha$ for every $n \in \omega$, by a Scholium of Fodor's Lemma to each f_n, we know there exists γ_n such that the set $S_n = f_n^{-1}(\gamma_n)$ is stationary, for each $n \in \omega$.

We claim at least one of the S_n's has a stationary complement. If not, each one contains a club, hence $\bigcap_{n\in\omega} S_n$ contains their intersection, which is also a club. By unboundedness it has some ordinal $\alpha > \sup_{n\in\omega} \gamma_n$, but then $x_{\alpha,n} = f_n(\alpha) = \gamma_n$ doesn't converge to α, a contradiction. ■

As another application, we will prove a really famous theorem[3] in combinatorial set theory, known as the Δ-System Lemma. In particular, it's really important when proving some Forcing Notions are in fact ccc. Let's first define a Δ-system.

Definition 4.4

A family of sets $\mathcal{A}$ forms a Δ-system with root R if, and only if, $X \cap Y = R$, whenever $X, Y \in \mathcal{A}$, with $X \neq Y$.

We also say $\mathcal{B}$ forms a Δ-system, if there is a set R such that $\mathcal{B}$ is a Δ-system with root R.

Corollary 4.2: Δ-System Lemma

Let $\mathcal{A}$ be a family of finite sets with $|\mathcal{A}| = \kappa$. Then there is a $\mathcal{B} \in [\mathcal{A}]^\kappa$ such that $\mathcal{B}$ forms a Δ-system.

Proof. We may assume WLOG that $\mathcal{A} \subseteq [\kappa]^{<\omega}$. Since every set in $\mathcal{A}$ is finite, and we have κ-many of those, by Pigeonhole we can find κ of them $\{A_\alpha : \alpha < \kappa\}$ of the same size $n \in \omega$. todo

[2] It's worth noting we can only prove it under the Axiom of Choice, it's consistent with ZF that every stationary set contains a club.

[3] Although it's called lemma, which we will in fact state as a corollary xD.

■

4.3 Silver's Theorem

Let's now prove the main theorem of this section. Roughly, it says that if GCH holds for all cardinals smaller than some singular with uncountable cofinality, then it also holds for it.

Theorem 4.2: Silver's Theorem

Let $\aleph_\lambda$ be a singular cardinal, with $\mathrm{cf}(\lambda) > \omega$. If $2^{\aleph_\alpha} = \aleph_{\alpha+1}$, for every $\alpha < \lambda$, then

$$2^{\aleph_\lambda} = \aleph_{\lambda+1}$$

Now, if λ itself is not singular,

different statement in the general case

In order to prove it, we will first prove a series of lemmas needed to show Silver's result.

Definition 4.5: Strong Limit Cardinal

A cardinal κ is called a strong limit cardinal if, for any $\lambda < \kappa$, we have $2^\lambda < \kappa$.

It's easy to see that, under GCH, the concept of a limit cardinal matches with the one of a strong limit[4] cardinal. Why we should care? Look again to Silver's Theorem hypothesis, we have some kind of partial GCH for all cardinal smaller than $\aleph_\lambda$. In fact, this imply that $\aleph_\alpha$ is a strong limit cardinal for every $\alpha < \lambda$. In particular, the next lemma holds.

Lemma 4.4

If $\aleph_\alpha$ is a strong limit cardinal, and if $\kappa, \lambda < \aleph_\alpha$ are infinite cardinals, then $\kappa^\lambda < \aleph_\alpha$.

Proof.

$$\kappa^\lambda \leq (\kappa \cdot \lambda)^{\kappa \cdot \lambda} = 2^{\kappa \cdot \lambda} < \aleph_\alpha$$

where in the inequality we used lemma 3.3. ■

Throughout this section, we will fix a λ as in Silver's Theorem, and let $\kappa = \mathrm{cf}(\lambda)$. Then, by the previous Lemma, we can conclude that

todo

Lemma 4.5

Let $\{A_\alpha : \alpha < \kappa\}$ be a family such that $|A_\alpha| \leq \aleph_\alpha$ for every $\alpha < \kappa$. If F is a family of almost disjoint choice functions for the family, then $|F| \leq \aleph_\kappa$.

Proof. We can assume WLOG that $A_\alpha \subseteq \omega_\alpha$. Let S_0 be the set of all limit ordinals between 0 and κ. Given $f \in F$ and $\alpha \in S_0$, let $f^*(\alpha)$ be the least β such that $f(\alpha) < \omega_\beta$.

Since $f(\alpha) \in A_\alpha \subseteq \omega_\alpha$, we have $f(\alpha) < \omega_\alpha$ and, since α is a limit ordinal, then $f(\alpha) < \omega_\beta$, for some $\beta < \alpha$. This implies that $f^*(\alpha) < \alpha$, i.e., it's a regressive function. Since S_0 is stationary, by

[4] yeah, this is the same strong limit we cited in the end of section 2.3

Fodor's Lemma there is some stationary $S \subseteq S_0$ such that $f^* \restriction_S$ is constant, let's say in β. Then $f \restriction_S$ is a function from $S \subseteq \kappa$ to $\omega_\beta < \omega_\kappa$. Note that both S and β depends on f.

Now, consider the function φ that takes some $f \in F$ and return exactly that $f \restriction_S : S \to \omega_\beta$. Then

$$\varphi : F \to \bigcup_{S \subseteq \kappa} \left(\bigcup_{\beta < \kappa} \omega_\beta^S \right)$$

We claim φ is injective. In fact, if $f \neq g$, even if $\varphi(f)$ and $\varphi(g)$ have the same domain S, since they're almost disjoint, there is some $\alpha < \kappa$ such that $f(\beta) \neq g(\beta)$ for every $\beta \geq \alpha$. But since S is stationary, there is some $\gamma > \alpha$ in it, so $\varphi(f)(\gamma) \neq \varphi(g)(\gamma)$. Then, since φ is injective,

$$|F| \leq \sum_{S \subseteq \kappa} \sum_{\beta < \kappa} \omega_\beta^{|S|} \leq 2^\kappa \sum_{\beta < \kappa} \omega_\beta^\kappa$$

■